Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW - 2

Date: 20-04-2026

Tech Fest 2026 – Online Event Ticket Booking System

SDG: SDG 8: Decent Work and Economic Growth

Presented By:

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INTRODUCTION

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- Traditional event coordination often suffers from manual booking errors and data inconsistencies.
- EventHub provides a centralized digital platform to automate ticketing and resource management.
- This project is crucial for enhancing transparency and reducing administrative overhead for organizers
- .
- It features a modern Spring Boot backend synchronized with a high-performance React frontend.
- The system offers real-world relevance through its application in festivals, seminars, and workshops.
- A dynamic seat mapping engine ensures efficient venue utilization and prevents overbooking.
- Advanced Admin Dashboards provide real-time insights into event performance and ticket sales.
- Overall, it serves as a comprehensive solution for modern, secure, and streamlined event management.

• PROBLEM STATEMENT

- Currently, many event management processes are still handled using manual methods or siloed spreadsheets
- This leads to inconsistencies in attendee data and significant delays in ticket processing.
- Organizers lack real-time visibility into ticket sales and event metrics across the platform.
- Fragmented systems make it difficult for users to discover and book diverse events seamlessly.
- There is a high risk of ticket duplication and fraud without a centralized digital verification system.
- Communication between organizers and attendees regarding event updates is often slow and inefficient.
- Therefore, a reliable, automated, and centralized system is needed to improve overall performance and accuracy.

OBJECTIVE(S)

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- **To design and develop** a centralized ticketing platform called "Tech Fest 2026" that automates event discovery and booking.
- **To implement** a dynamic seat management system that ensures real-time availability and prevents overbooking.

PROPOSED SYSTEM

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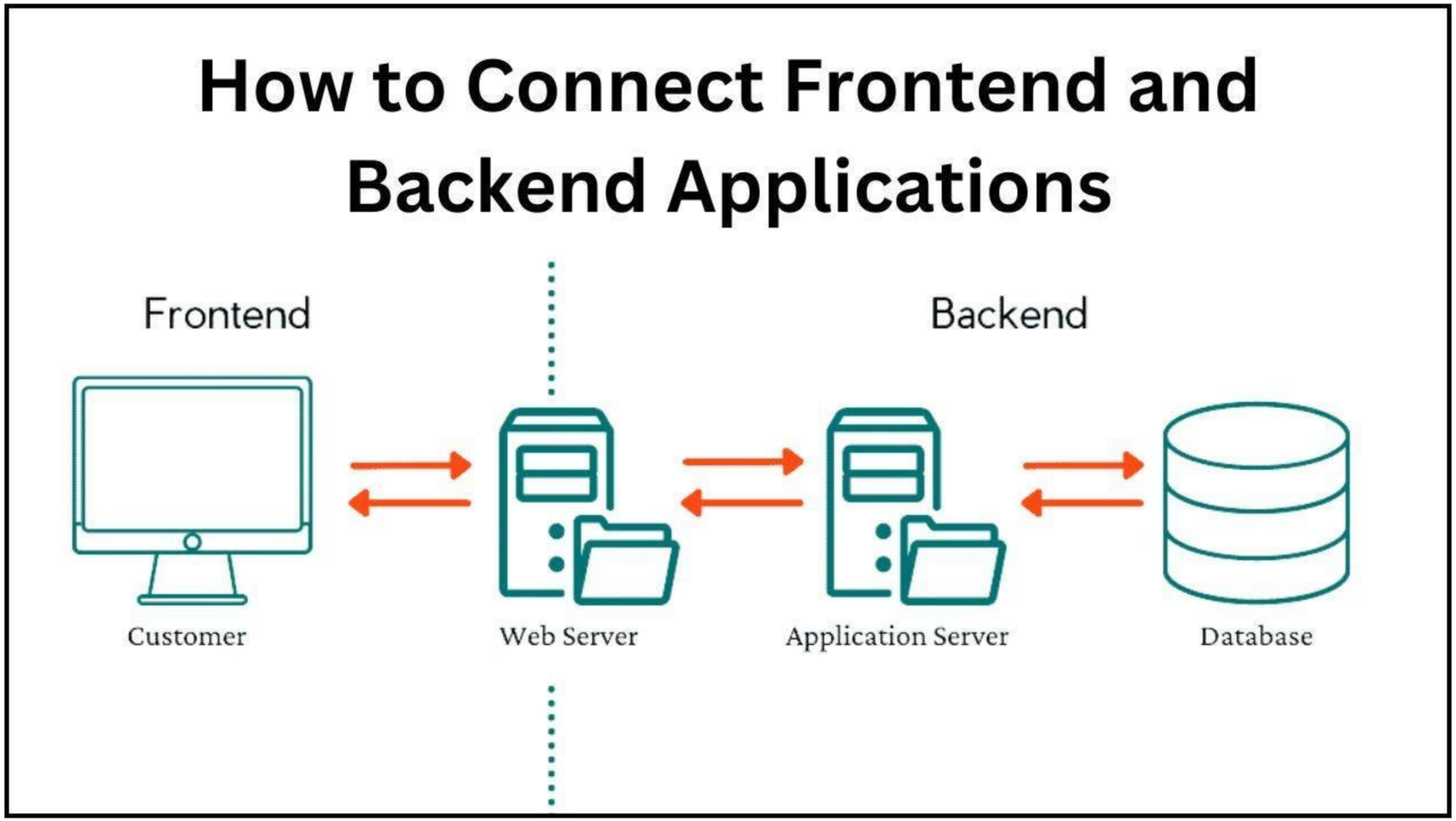
- Description: A full-stack web application leveraging Spring Boot for the backend and React for a modern, responsive frontend.
- Real-time Booking: Users can instantly browse available events and purchase tickets with live updates.
- Automated Seat Mapping: Implements a dynamic grid system (A, B... AA, AB) that handles varying venue capacities.
- Admin Dashboard: Provides powerful management tools for organizers to track sales and monitor attendee lists.
- Centralized Database: Uses H2/SQL to maintain a single source of truth for all events and transaction records.
- Secure Authentication: Features role-based access control to distinguish between attendees and administrators.
- Seamless UI/UX: Designed for a "premium" feel with smooth transitions, dark mode, and mobile-responsive layouts.
- Scalability: The modular architecture allows for the easy addition of new features and event categories.

PROPOSED SYSTEM ARCHITECTURE



WORKFLOW

EventHub **Ticket Booking Workflow**

TECHNOLOGIES USED

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Frontend: React.js (v19), Tailwind CSS.

Backend: Java 17, Spring Boot 3.4.0, Maven, Spring Data JPA (Hibernate).

Database: H2 (High-performance, in-memory/file-based relational database).

Tools: VS Code (IDE), ECLIPSE.

Software Requirements: Windows 10/11, Node.js v20+, Java JDK 17, Modern Web Browser.

Hardware Requirements: 8GB RAM minimum, Intel Core i5 Processor or equivalent, 500MB Free Disk Space.

MODULE DESCRIPTION

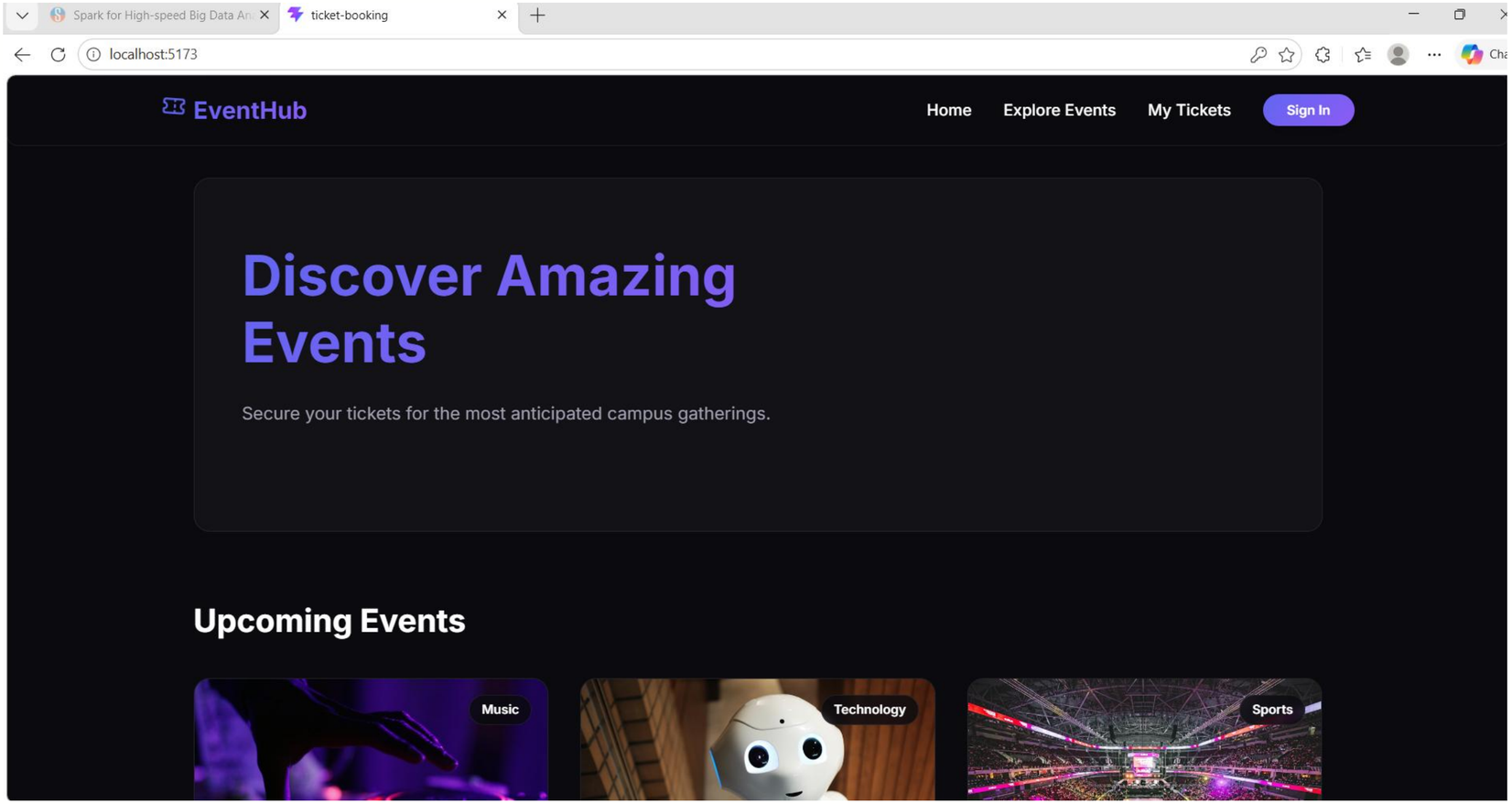
Module 1: User Authentication & Profiles: Handles secure user registration and login, maintaining persistent sessions and distinguishing between regular users and administrative staff.

Module 2: Event Discovery & Grid Management: Implements the dynamic event showcase and the advanced seat selection engine featuring robust, automated row labeling.

Module 3: Transactional Booking Engine: Manages the core business logic for ticket reservations, ensuring real-time seat synchronization and preventing overbooking in the database.

Module 4: Admin Management Console: A dedicated module for platform administrators to create events, update ticket capacities, and monitor global booking statistics.

IMPLEMENTATION SCREENSHOTS



A sleek Tech Fest 2026 homepage showcasing upcoming campus events with a modern dark-themed interface and vibrant visuals. The platform highlights categories like Music, Technology, and Sports, making ticket booking simple and engaging.

IMPLEMENTATION SCREENSHOTS

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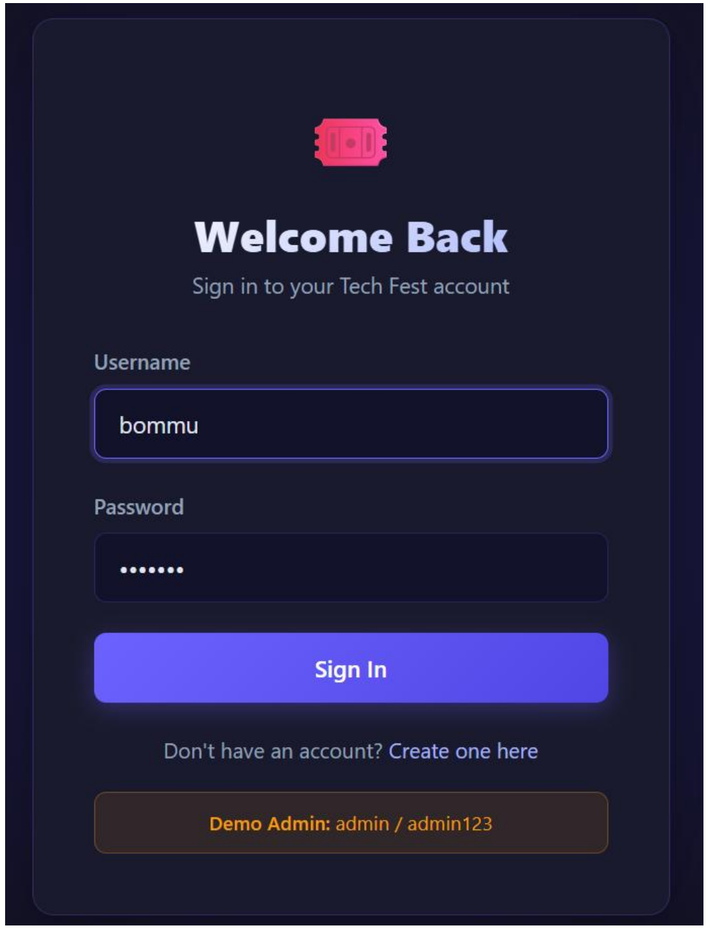
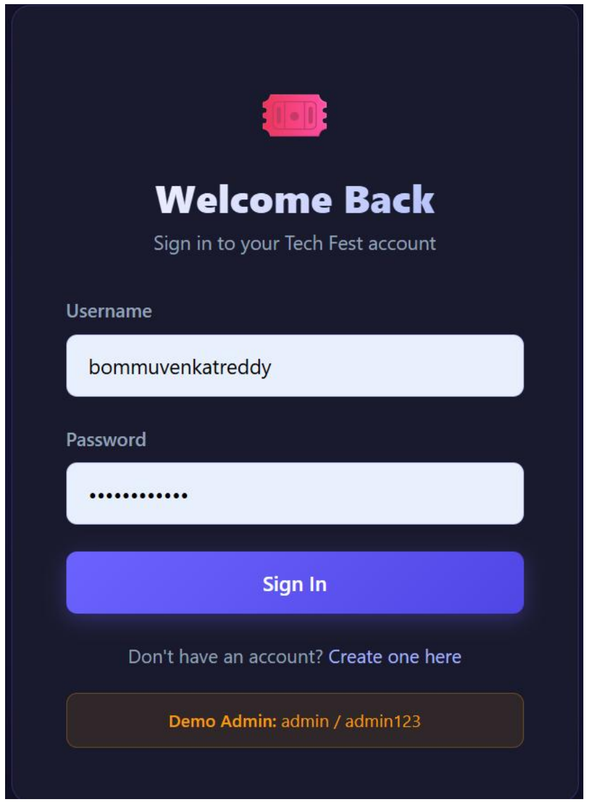
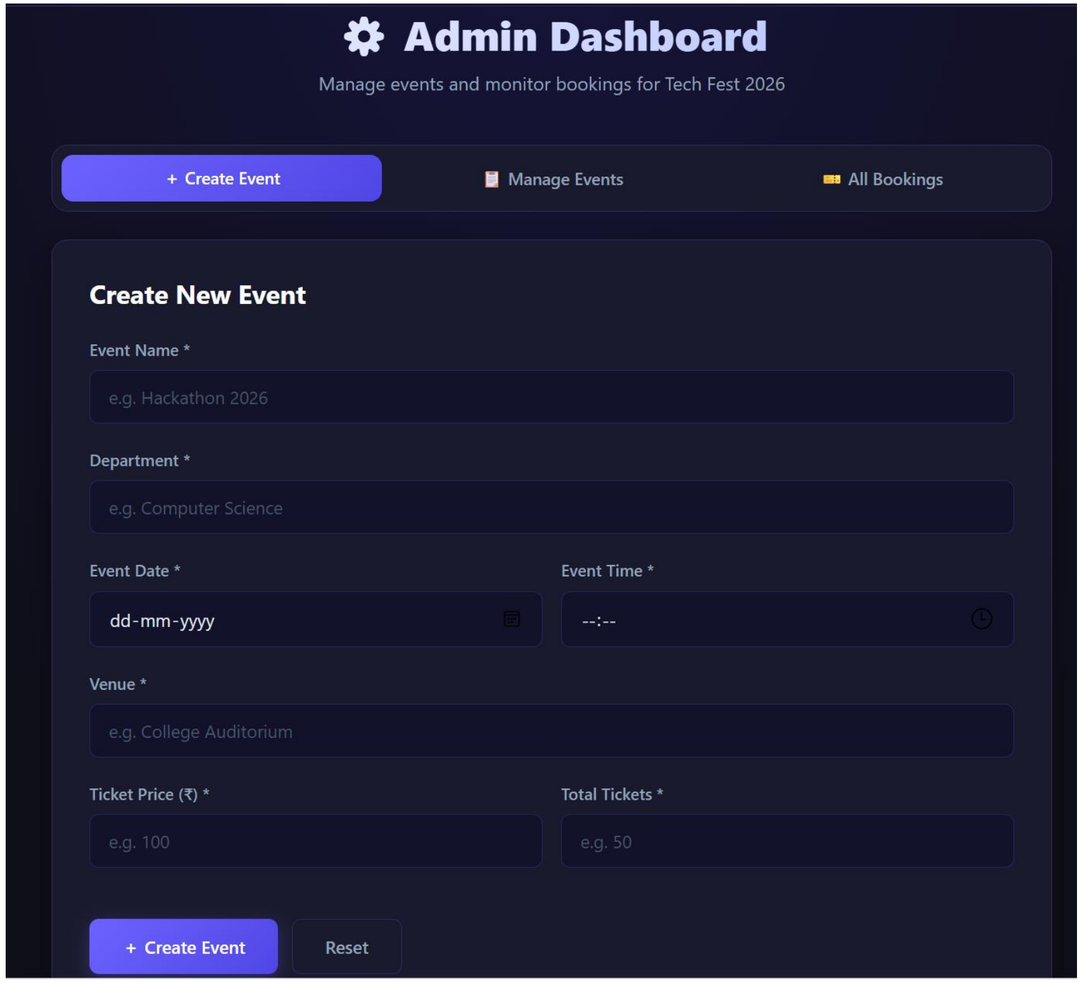




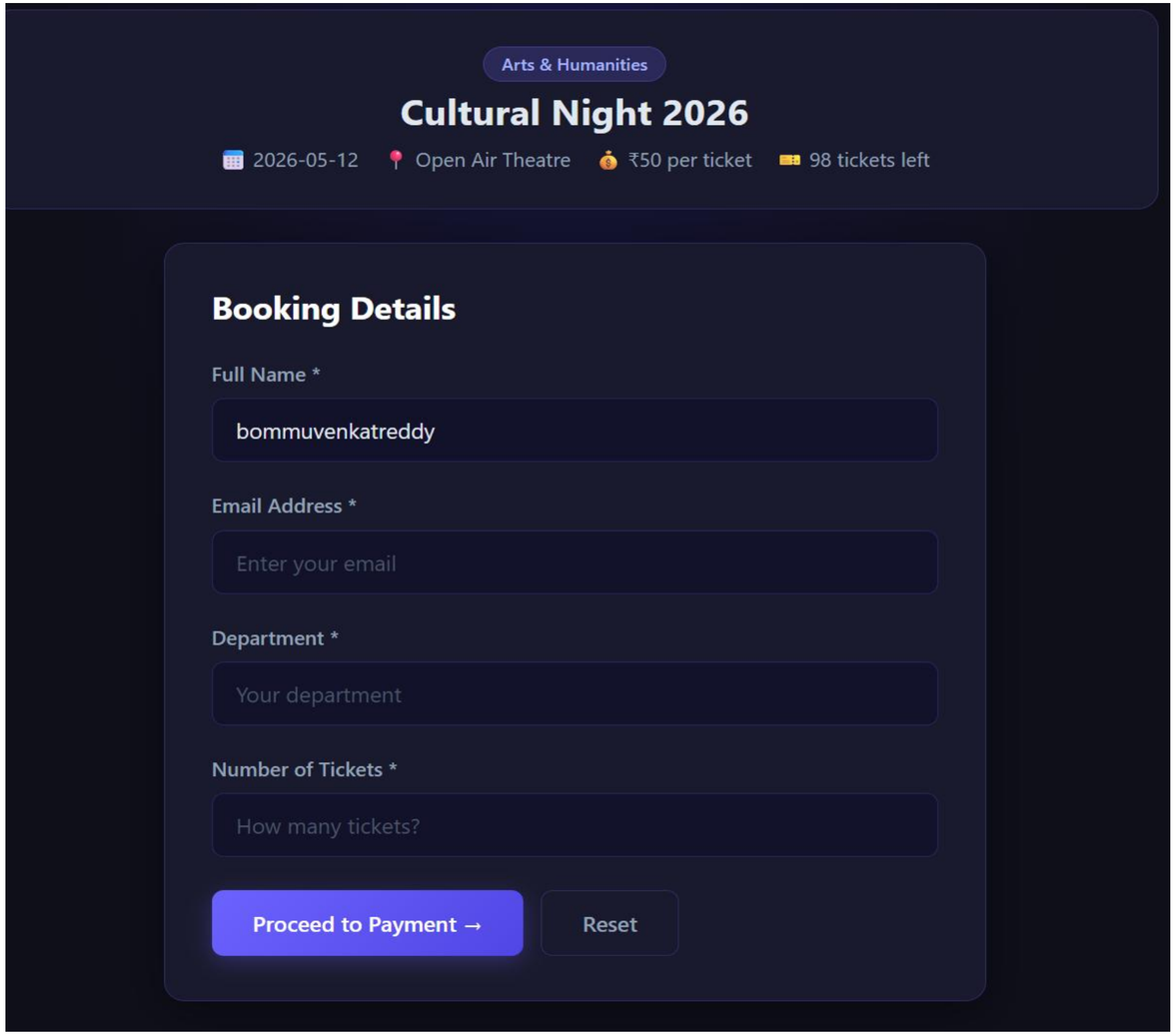
A secure and modern admin login portal for Tech Fest 2026, designed with a sleek dark theme and authentication interface. It enables administrators to manage events, user bookings, and platform operations through a centralized control center.

IMPLEMENTATION SCREENSHOTS



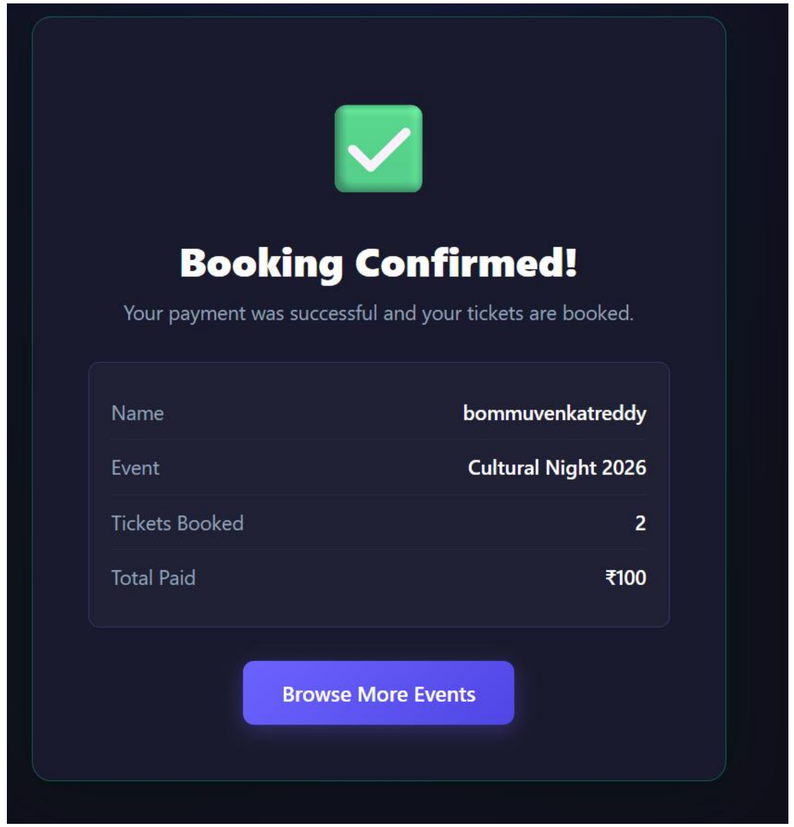
A dynamic Tech Fest 2026 admin dashboard that allows organizers to add, manage, and monitor campus events efficiently. It features event creation forms and a real-time active events panel for seamless administration.

IMPLEMENTATION SCREENSHOTS



An interactive Tech Fest 2026 ticket booking page displaying event details, seat selection, and user registration in a modern interface. Users can choose available seats, enter booking information, and confirm tickets seamlessly for live events.

IMPLEMENTATION SCREENSHOTS



A live backend bookings dashboard displaying real-time ticket reservations, user details, event names, and payment totals. It helps administrators track bookings efficiently and monitor platform transactions instantly.

ADVANTAGES & LIMITATIONS

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ADVANTAGES

- Complete Automation
- Real-time Synchronization
- Enhanced User Experience
- Centralized Data Management
- High Scalability
- Transparency and Trust

DISADVANTAGES (LIMITATIONS)

- Internet Dependency
- Infrastructure Requirements
- Local Storage Limitations
- Payment Gateway Complexity
- Technical Maintenance
- Learning Curve.

Thank you